



**MERIDIAN JUNIOR COLLEGE**  
JC2 Preliminary Examinations 2013  
Higher 1

CANDIDATE  
NAME

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CIVICS  
GROUP

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INDEX  
NUMBER

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## H1 BIOLOGY

**8875/02**

Paper 2

**18 September 2013**

**2 hours**

Additional Materials: Answer papers

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### READ THESE INSTRUCTIONS FIRST

**Do not open this booklet until you are told to do so.**

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

#### Section A

Answer **all** questions.

#### Section B

Answer any **one** question.

At the end of the examination,

1. Fasten your answer papers to section B securely together.
2. Hand in the following separately:
  - Section A
  - Section B

The number of marks is given in brackets [ ] at the end of each question or part question.

| For examiner's Use |      |
|--------------------|------|
| Section A          |      |
| 1                  | / 9  |
| 2                  | / 10 |
| 3                  | / 10 |
| 4                  | / 11 |
| Section B          |      |
| 5 / 6              | / 20 |
| Total              | / 60 |

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This paper consists of **13** printed pages.

**[Turn over]**

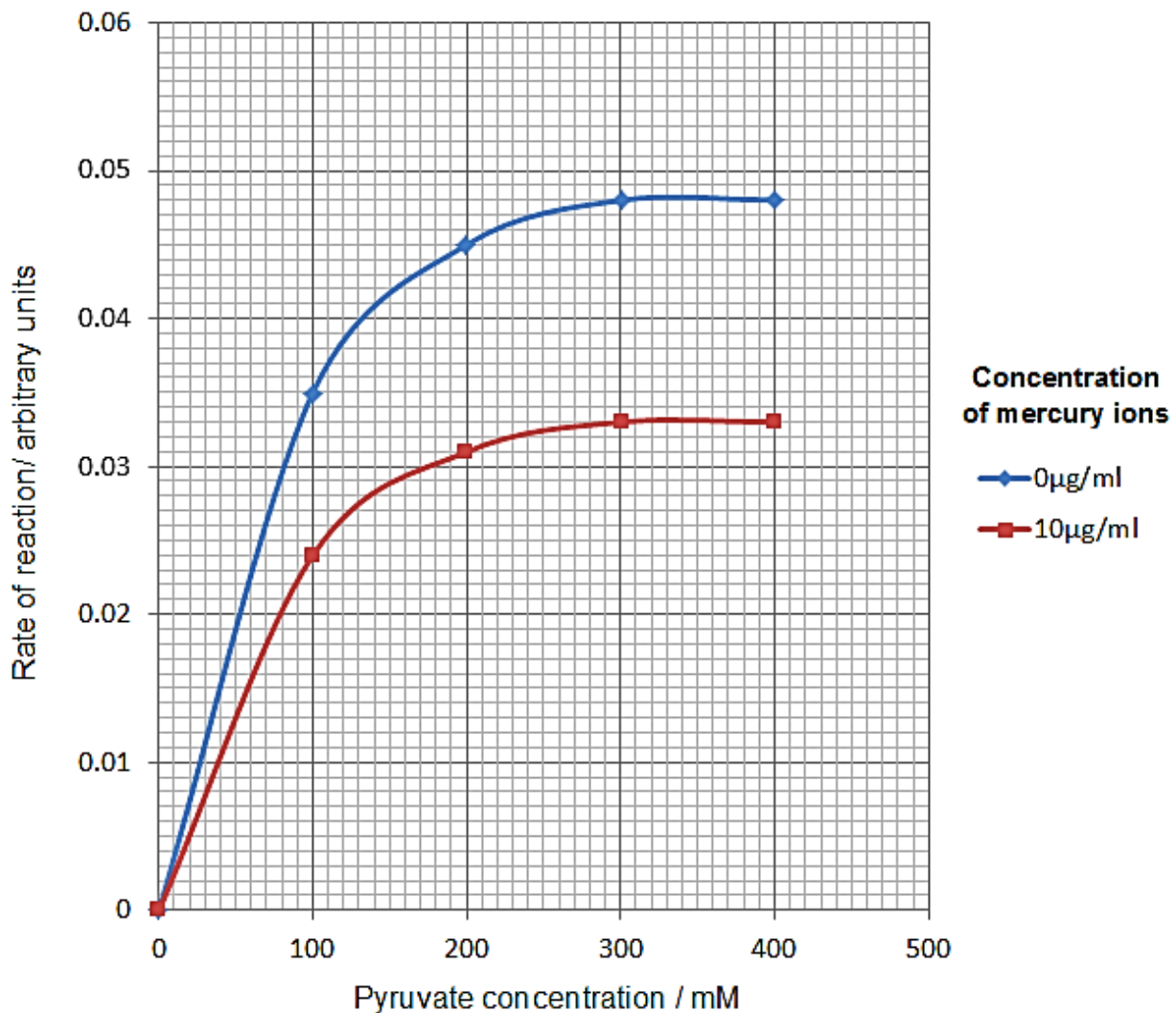
**Section A**  
Answer **all** the questions in this section.

For  
Examiner's  
Use

**QUESTION 1**

Heavy metal ions are waste products of many industrial processes and at times may leak into the ocean and cause environmental pollution. The toxicity of heavy metals on organisms stems from their effect on metabolically-important enzymes, such as lactate dehydrogenase. Lactate dehydrogenase catalyses the conversion of pyruvate into lactate.

Fig. 1.1 shows the effect of varying concentrations of mercury ions on the rate of reaction catalysed by lactate dehydrogenase isolated from *Oreochromis mossambicus*, a species of fish.



**Fig. 1.1**

- (a)** Describe and explain the effect of mercury ions on the rate of reaction catalysed by lactate dehydrogenase. [4]

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- (b)** Explain how mercury ions may affect the survival of the fish when oxygen concentrations are low. [3]

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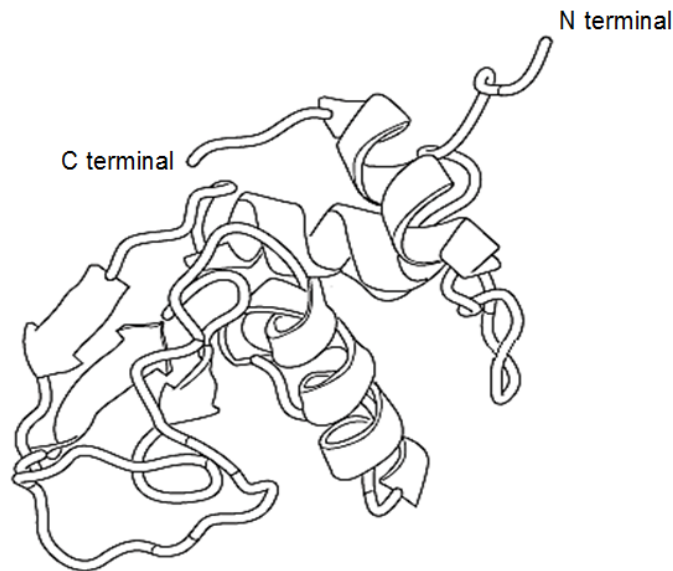
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Human lysozyme is an important enzyme which is part of the human body's defence against some bacteria. It is found in abundance in tears, saliva and mucous. The main action of lysozyme is to catalyse the hydrolysis of  $\beta$ -(1,4) glycosidic bonds between N-acetylmuramate (NAM) and N-acetylglucosamine (NAG) sugar residues in the peptidoglycan cell walls of bacteria. With the weakened cell wall, the bacterial cells are more likely to undergo lysis.

Fig. 1.2 shows the structure of lysozyme



**Fig. 1.2**

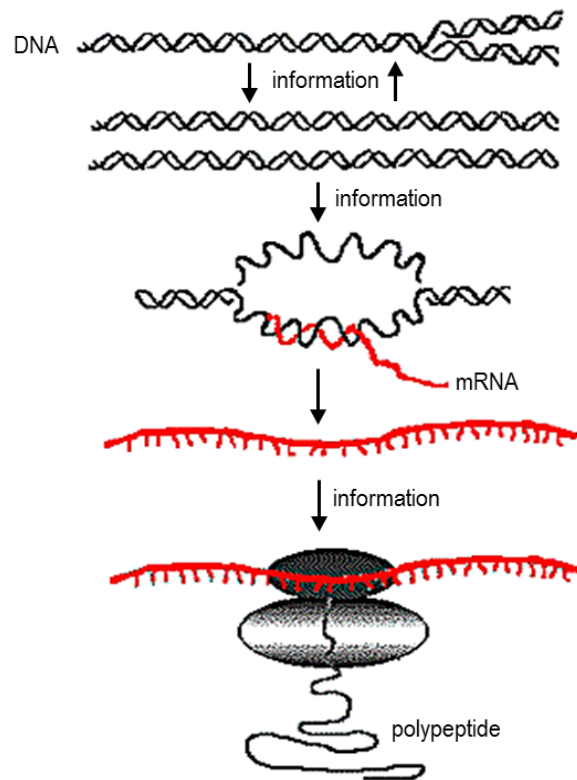
(c) With reference to Fig. 1.2 and with your knowledge on haemoglobin, state two structural differences between lysozyme and haemoglobin. [2]

1. ....
- .....
2. ....
- .....

**[Total: 9]**

## QUESTION 2

Fig. 2.1 illustrates the Central Dogma, which describes the flow of genetic information of DNA to RNA to proteins.



**Fig. 2.1**

- (a) Describe how the information carried on mRNA is related to its role in translation in a prokaryotic cell. [3]

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A ribonucleoprotein is a protein that is associated with RNA. The ribosome is an example of a ribonucleoprotein. Prokaryotes have 70S ribosomes while eukaryotes have 80S ribosomes, each consisting of a small and large subunit. In prokaryotes, the small subunit is made up of a 16S rRNA associated with 21 proteins. Fig. 2.2 shows the structure of 16S rRNA.

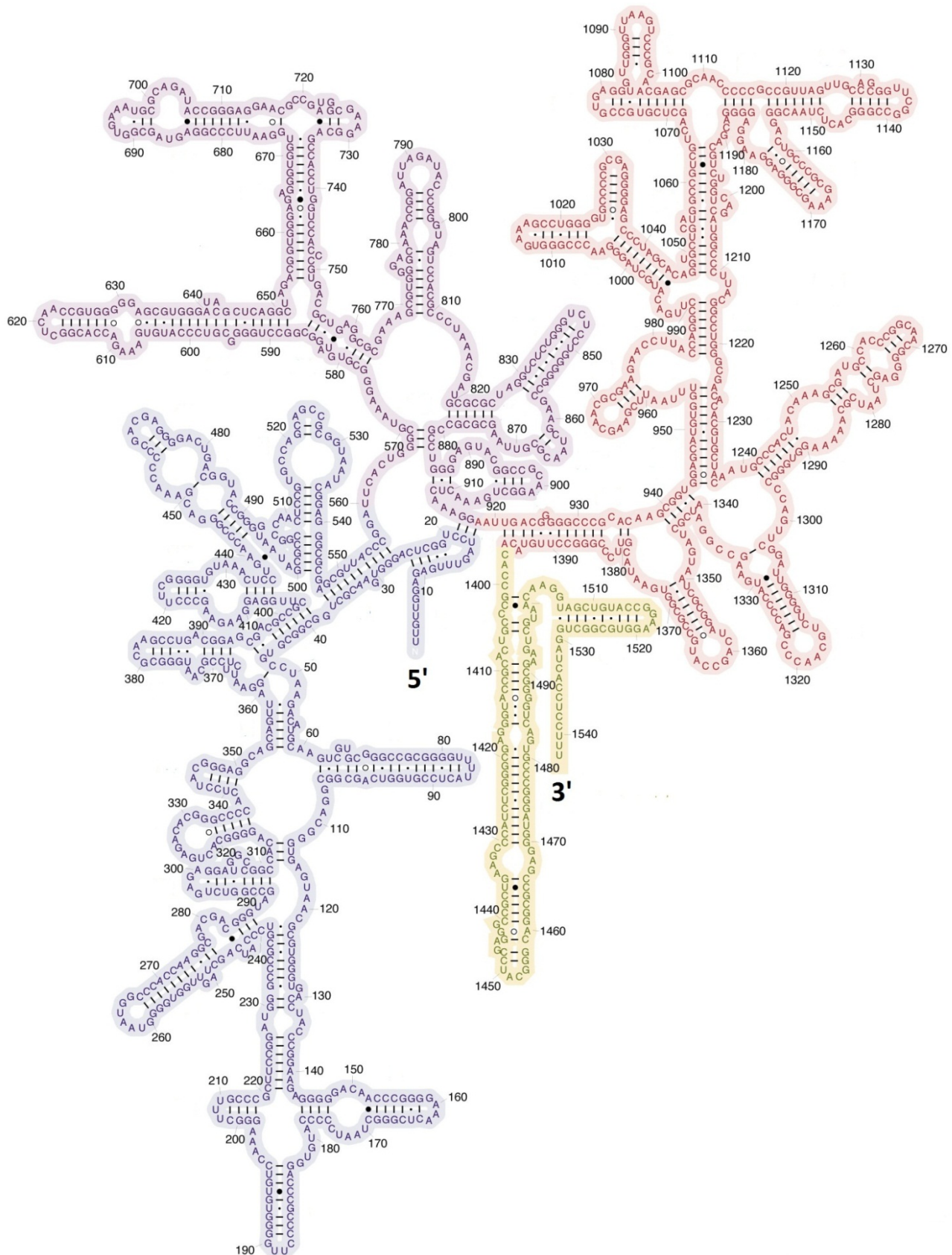


Fig. 2.2

- (b)** 16S rRNA and tRNA are both made up of four types of ribonucleotides linked by phosphodiester bonds.

State two other similarities between the structures of the 16S rRNA and tRNA. [2]

1. ....  
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2. ....  
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- (c)** Describe how eukaryotic ribosomes are formed. [4]

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Unlike eukaryotes, prokaryotes do not have endoplasmic reticulum or vesicles.

- (d)** Suggest how prokaryotes are still able to secrete proteins. [1]

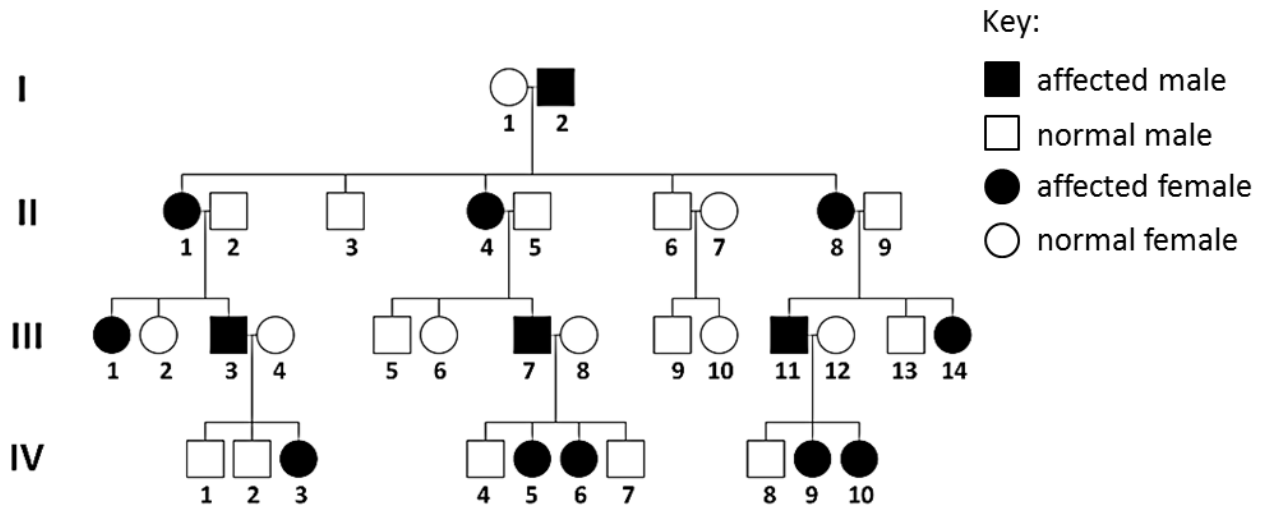
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**[Total: 10]**

### QUESTION 3

Hereditary hypophosphatemic rickets is a genetic disorder that results on low level of phosphate in the blood (hypophosphatemia).

Fig. 3.1 shows the inheritance of this disease over four generations in an extended family.



**Fig. 3.1**

(a) Explain the mode of inheritance of this disease.

[2]

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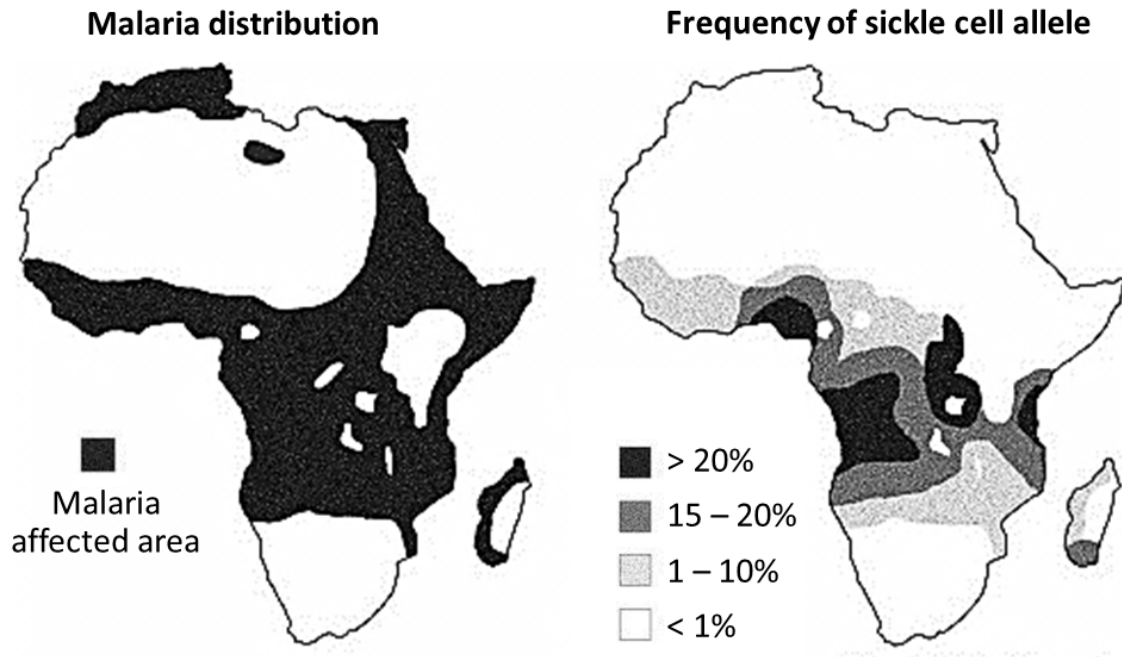
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(b) Using the symbols **D/d** to represent the alleles, construct a genetic diagram to show how individual **III-7** and **III-8** can pass on the disease to their children.

[4]



Fig. 3.2 shows the distribution of malaria and the allele frequency of the sickle cell allele.



**Fig. 3.2**

- (a) Describe and explain the correlation between the occurrence of malaria and the frequency of the sickle cell allele. [4]

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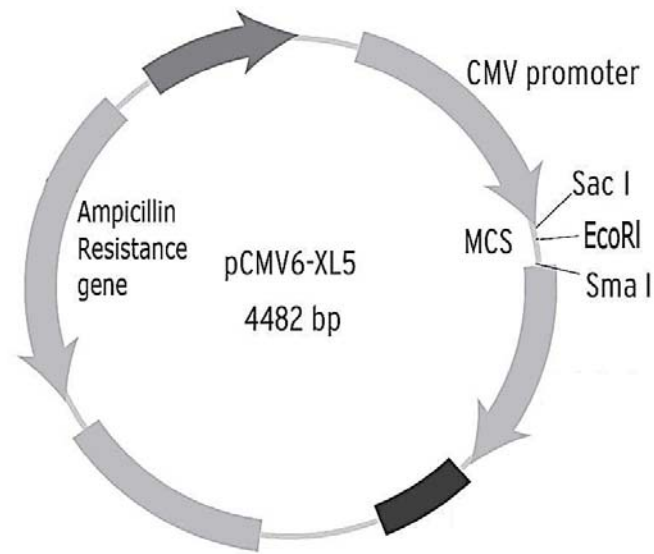
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**[Total: 10]**

**QUESTION 4**

pCMV6-XL5 is a plasmid with a multiple cloning site (MCS) that lies downstream of the CMV promoter (Fig. 4.1). This plasmid can be inserted into both eukaryotic and prokaryotic host cells. The arrows denote the direction in which the genes are transcribed.



**Fig. 4.1**

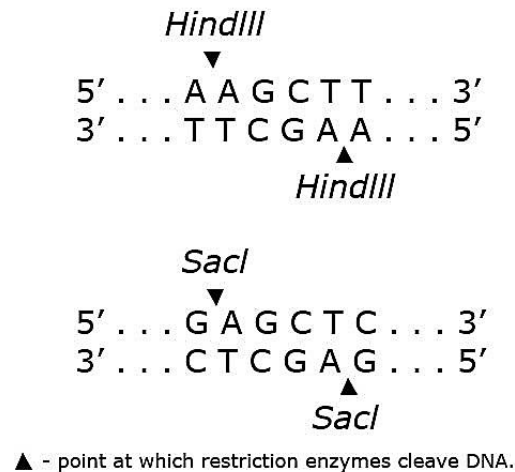
**(a)** State the function of the CMV promoter in pCMV6-XL5.

[1]

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An artificially-synthesised human growth hormone (hGH) gene with flanking *Hind*III restriction site sequences was created. The restriction sites for the restriction enzymes *Hind*III and *Sac*I are shown in Fig. 4.2.



**Fig. 4.2**

- (b) Using information in Fig. 4.1 and 4.2, explain how the hGH gene can be inserted into pCMV6-XL5. [3]

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- (c) Describe the steps required to select bacterial colonies containing the human growth hormone gene after artificial transformation has taken place. [5]

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- (d)** Plasmid replication is usually initiated by RNAI, which is a sequence of ribonucleotides that binds to plasmid DNA prior to replication.

State and explain the role of RNAI in plasmid replication.

[2]

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**[Total: 11]**

**Section B**  
Answer **ONE** question

For  
Examiner's  
Use

Write your answers on the separate answer paper provided.  
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.  
Your answers must be in continuous prose, where appropriate.  
Your answers must be set out in questions **(a)**, **(b)**, etc., as indicated in the question.

**QUESTION 5**

- (a)** Explain how genetically modified organisms may provide a solution to global food shortage issues. [8]
- (b)** Explain how reduced NADP is formed in plants cell and how it is used in photosynthesis. [7]
- (c)** Explain how molecular homology supports Darwin's theory of natural selection. [5]

**[Total: 20]**

**QUESTION 6**

- (a)** Describe and explain how the structures of chloroplasts and mitochondria are similarly adapted for their functions [8]
- (b)** Compare the features of zygotic stem cells and adult stem cells. [6]
- (c)** Discuss the goals of the Human Genome Project. [6]

**[Total: 20]**

**• End of Paper •**